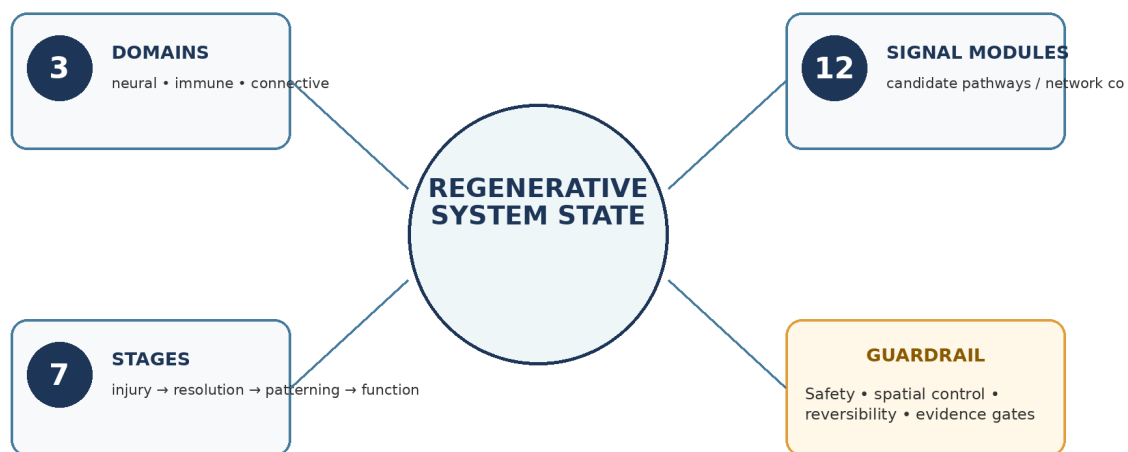


ANIMAL-INSPIRED REGENERATIVE SYSTEMS

A Conceptual 3–7–12 Framework for Limb Regeneration Research

From comparative regeneration biology to a testable, safety-gated translational research program

Figure 2. The proposed 3–7–12 systems framework



Conceptual architecture of the proposed framework (original figure; hypothesis map, not clinical evidence).

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Version: August 2026 • Revised conceptual manuscript

Scientific integrity and scope

This revised manuscript deliberately converts the earlier draft from a claim of completed experimental validation into a clearly bounded conceptual and translational research proposal. The supplied draft reported a rat experiment, numerical regeneration outcomes, a 3–7–12 simulation, and several highly specific 2025–2026 citations. Those elements are not independently documented in the supplied file by raw datasets, protocols, ethics approvals, preregistration, or an auditable code repository. They are therefore not presented here as established experimental findings.

The central contribution retained from the original manuscript is the author’s 3–7–12 organizing framework: three interacting biological domains, seven regenerative process stages, and a set of twelve candidate signalling modules. In this version, the framework is explicitly identified as a conceptual hypothesis-generating architecture rather than a discovered biological law.

This distinction is important for scientific credibility, corporate due diligence, and responsible investment. A strong translational paper should separate established biology, author-generated hypotheses, proposed experiments, and future commercial opportunities. The revised manuscript follows that structure.

Executive positioning

Element	Status in this revision
Comparative animal regeneration	Evidence-informed literature synthesis
3–7–12 framework	Author-proposed conceptual systems model
Human limb regeneration	Long-term research objective; not demonstrated
Rat regeneration results in the earlier draft	Removed as an unsupported completed experiment
Clinical protocol	Not proposed as a treatment; only research directions
Commercial proposition	Research platform / IP and collaboration opportunity
Investment status	Preclinical concept; investment subject to independent diligence

Abstract

Complex tissue regeneration requires more than cell proliferation. Regenerative organisms coordinate injury sensing, immune resolution, extracellular-matrix remodeling, positional information, vascular support, neural interactions, and growth termination in space and time. Recent work in axolotl limb regeneration has clarified the importance of positional memory, connective-tissue signalling centres, retinoic-acid regulation, Hedgehog-dependent patterning, and vascular signalling. At the same time, human translation remains constrained by the limited regenerative competence of adult mammals, safety requirements for cell-based interventions, and the difficulty of reproducing large-scale pattern formation.

This manuscript proposes the 3–7–12 framework as a systems-level hypothesis generator for regenerative medicine. Code 3 organizes regeneration around three coupled domains—neural, immune, and connective/structural. Code 7 represents a process sequence from injury and haemostasis through inflammatory resolution, progenitor activation, pattern formation, tissue growth, maturation, and functional integration. Code 12 is used as a candidate network layer covering twelve signalling families frequently implicated in regeneration and tissue repair; it is not asserted that exactly twelve pathways universally govern regeneration.

The framework is deliberately designed for falsifiability. It proposes computational modelling, human-cell and organoid experiments, spatial transcriptomics/proteomics, and staged animal studies to test whether coordinated multi-domain control can improve tissue repair without inducing uncontrolled proliferation, fibrosis, ectopic tissue, or systemic toxicity. The manuscript also defines evidence gates, intellectual-property opportunities, partnership models, and a commercialization pathway. The resulting proposition is not a ready-to-use clinical treatment; it is a research platform intended to help laboratories, biotechnology companies, medical-device developers, and strategic investors evaluate a coherent route from comparative biology to regulated translational development.

Keywords: regenerative medicine; limb regeneration; axolotl; positional memory; tissue engineering; systems biology; computational modelling; 3–7–12 framework; translational research; regenerative biotechnology.

1. Introduction: from biological inspiration to testable engineering

Adult mammals generally repair injuries through inflammation, matrix deposition, vascular remodelling and scar formation rather than recreating a complete anatomical structure. By contrast, salamanders such as the axolotl can regenerate complex appendages. Modern regeneration research increasingly treats this difference as a systems problem: regenerative competence emerges from the interaction of cell states, positional information, signalling gradients, immune responses, nerves, vasculature, extracellular matrix and growth control.

Recent axolotl studies are particularly relevant. Work published in 2025 showed that connective-tissue cells retain positional information and can form signalling centres after amputation, while another study demonstrated that retinoic-acid degradation helps determine proximodistal identity and identified Shox as an RA-responsive component. These findings support the idea that regeneration is not simply 'more stem cells'; it is a spatially encoded reconstruction process.

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A second translational lesson is that individual pathways are unlikely to be sufficient. For example, recent work found that VEGF signalling contributes to blastema growth beyond its familiar vascular role, while studies of axolotl digits indicate that Hedgehog activity can be important for joint regeneration but is not by itself sufficient to restore regeneration fidelity.

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The central question of this paper is therefore not 'which single gene should be switched on?' but: can a controlled, spatially explicit, multi-domain regulatory architecture reproduce selected properties of regenerative biology in mammalian tissues?

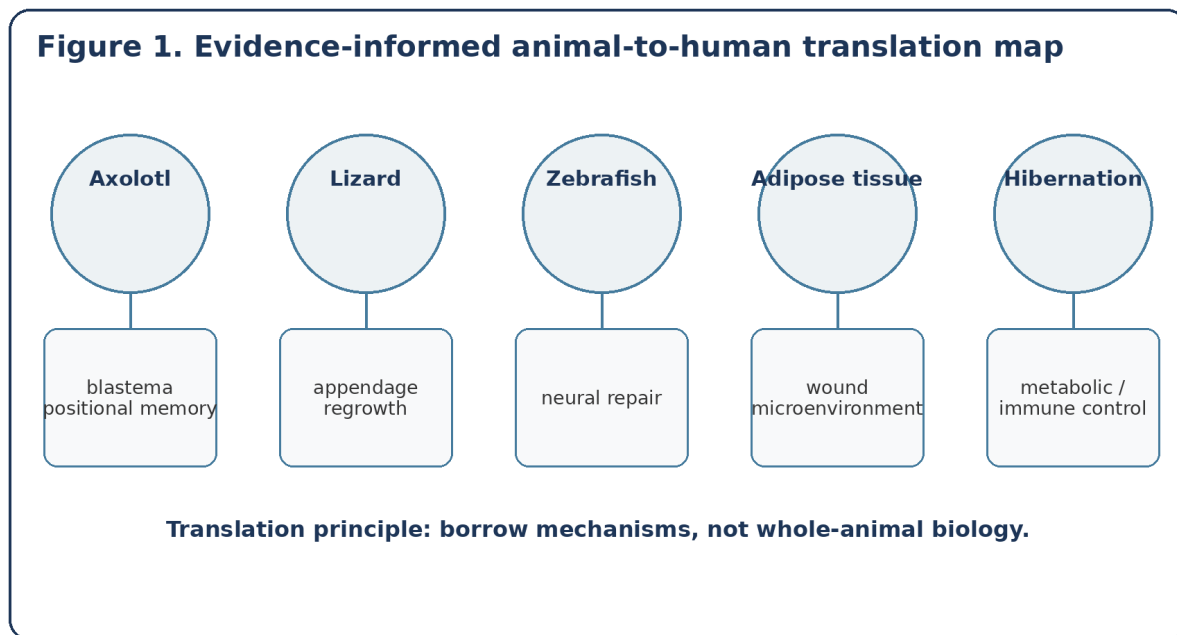


Figure 1. Evidence-informed mapping from model-organism mechanisms to translational research questions. Original synthesis figure; it does not imply that the listed mechanisms have already been translated into human limb regeneration.

2. Comparative regeneration: seven biological lessons

Model / system	Evidence-supported lesson	Translational question
Axolotl	Blastema formation and positional memory enable reconstruction of missing structures.	Can positional information be preserved or engineered in mammalian repair tissues?
Lizard appendage	Regeneration can be spatially restricted within one organism.	What local tissue conditions distinguish regenerative from non-regenerative responses?
Zebrafish	Certain neural tissues show substantially stronger regenerative responses than adult mammals.	Which pro-regenerative glial and immune programs are conserved?
Adipose-derived cells	Human studies support potential benefit in selected wound-healing settings, while standardization and long-term evidence remain challenges.	Can a controlled cell-derived microenvironment support structural repair without fibrosis?
Hibernating mammals	Metabolic and immune adaptations provide hypotheses about injury tolerance and inflammation control.	Can transient metabolic or inflammatory modulation improve repair safely?
Turtle / stress biology	Stress-response systems illustrate how cells can survive extreme conditions.	Can stress protection improve survival of engineered tissues or transplanted cells?
Cross-species principle	Regeneration depends on coordinated signals, spatial context and termination of growth.	Can these features be formalized as an engineering control problem?

The literature does not justify treating these animals as interchangeable 'keys' to human regeneration. Each model captures only a subset of regenerative biology. The value of comparative biology is therefore combinatorial and hypothesis-generating rather than a direct recipe for clinical treatment.

2.1 Axolotl: positional memory as a central engineering problem

Axolotl connective-tissue cells retain positional information that helps regenerating tissues reconstruct the appropriate segment. The 2025 Nature study described distinct signalling centres and demonstrated that modifying positional memory changes signalling outputs. A companion Nature Communications study showed that retinoic-acid breakdown through CYP26B1 helps establish proximodistal identity and that Shox responds to RA-dependent positional information. [NciteÖturn0search4Öturn0search1Ö](#)

2.2 Vascular and growth support

VEGF is often introduced as an angiogenesis factor, but recent axolotl work indicates a broader role in blastema growth and proliferation. This supports a systems interpretation in which vascular signalling is coupled to tissue growth rather than treated as an independent module. [NciteÖturn1search9Ö](#)

2.3 Neural and immune context

Regenerative species provide evidence that neural, glial and immune responses can be coordinated with tissue reconstruction. Reviews of zebrafish spinal-cord regeneration emphasize the interaction of pro-regenerative glial, neuronal and immune responses. [NciteÖturn0search13Ô](#)

3. The 3–7–12 framework: a systems hypothesis

The 3–7–12 framework is proposed here as an organizing language, not as a claim that nature contains a universal numerical code. Its scientific value can only be established through predictive tests: the framework should generate hypotheses that outperform simpler models, identify measurable failure modes, and remain useful when the data disagree with the initial assumptions.

Figure 2. The proposed 3–7–12 systems framework

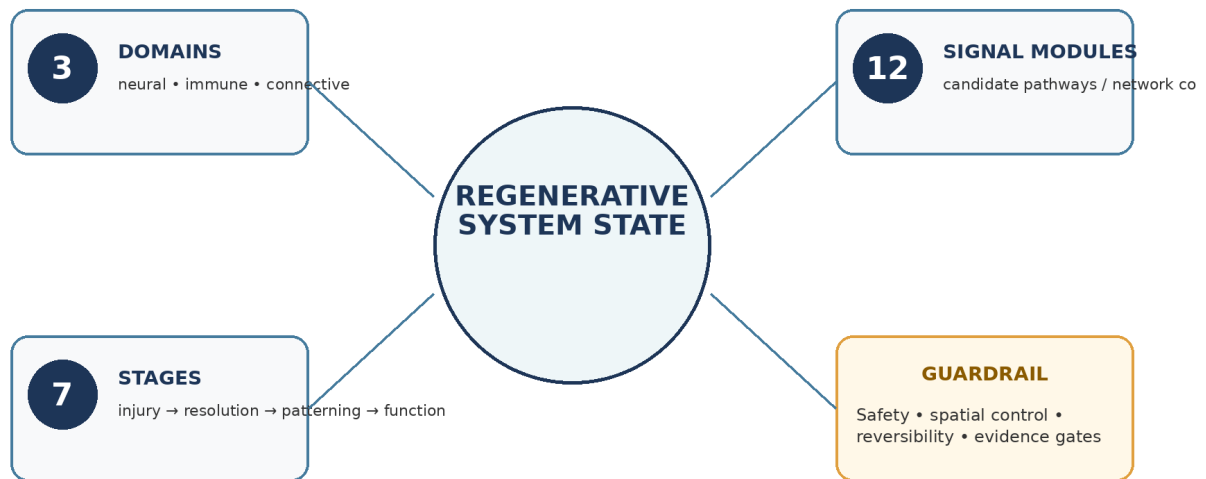


Figure 2. Proposed 3–7–12 architecture. The numbers are organizational components of the author's framework, not established biological constants.

3.1 Code 3 — three coupled domains

Domain A: neural/positional signalling. This includes nerve-associated signals, neural progenitor states and spatial patterning information. Domain B: immune regulation. This covers injury sensing, inflammatory activation, resolution, macrophage states and immunometabolic context. Domain C: connective/structural tissue. This includes fibroblasts, progenitors, extracellular matrix, vascular support and mechanical properties.

The hypothesis is that a regenerative response is more stable when these domains are controlled jointly. The model should therefore measure all three rather than using a single endpoint such as wound closure.

3.2 Code 7 — seven functional stages

The proposed seven-stage sequence is: (1) injury containment and haemostasis; (2) inflammatory activation; (3) transition to regenerative inflammation; (4) progenitor/niche activation; (5) positional patterning and tissue specification; (6) tissue growth, vascularization and structural integration; and (7) maturation, functional recovery and growth termination.

These stages are deliberately broader than the earlier manuscript's rigid blastema sequence. They are intended as measurable states that can overlap in time rather than seven perfectly discrete biological boxes.

3.3 Code 12 — candidate signalling network

The initial network includes Hedgehog, Wnt, FGF, TGF- β /BMP, retinoic-acid signalling, Notch, JAK–STAT, Hippo, PI3K–AKT, MAPK/ERK, NF- κ B and VEGF. These are candidate modules, not a claim that exactly twelve pathways are universally active in every regenerative species or tissue.

Module	Primary hypothesis role
Hedgehog	Patterning / progenitor and joint-related signalling
Wnt	Cell state, proliferation and tissue patterning
FGF	Growth and developmental signalling
TGF- β /BMP	Matrix, differentiation and skeletal signalling
Retinoic acid	Positional identity and patterning
Notch	Cell fate and boundary control
JAK–STAT	Cytokine and immune-state coupling
Hippo	Growth control and tissue-size regulation
PI3K–AKT	Survival and growth
MAPK/ERK	Proliferation and differentiation
NF- κ B	Inflammatory response
VEGF	Vascular and blastema growth support

4. Core hypothesis and falsifiable predictions

Core hypothesis: a spatially and temporally controlled combination of regenerative microenvironmental signals can improve structural repair in mammalian tissue compared with single-factor interventions, provided that patterning and growth termination are explicitly controlled.

Prediction	Testable statement
P1	Multi-domain control will outperform single-domain control on composite repair metrics.
P2	Preserving positional information will improve anatomical fidelity, not merely tissue volume.
P3	Early inflammatory modulation will reduce fibrosis only when followed by appropriate regenerative signalling.
P4	Vascular support will influence both perfusion and regenerative tissue expansion.
P5	A control layer that actively terminates growth will reduce ectopic tissue and tumour-like overgrowth risk.
P6	Computational models can identify combinations that are robust to biological variability before large animal studies.

4.1 What would falsify the framework?

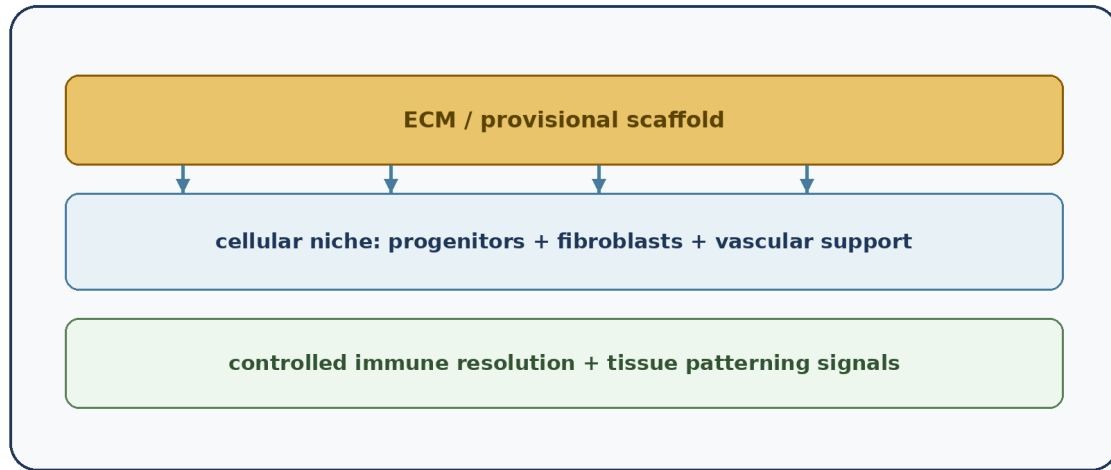
The framework should be considered unsuccessful if coordinated interventions do not outperform simpler controls; if gains in tissue volume do not translate into structural organization; if positional fidelity cannot be improved; if safety events increase disproportionately; or if model predictions fail to replicate in independent experiments. Explicit failure criteria make the project more credible to scientific and industrial partners.

5. Research program: from cells to integrated tissues

The revised program is intentionally staged. No step below should be interpreted as a clinical treatment recommendation. The purpose is to establish evidence before escalation.

Stage	Workstream	Focus	Evidence gate
Stage 1	Literature + data integration	Curate regeneration datasets; define measurable states; map conserved vs species-specific mechanisms.	Deliverable: auditable evidence map + model specification.
Stage 2	Human-cell / organoid studies	Test candidate microenvironmental modules individually and in combinations under controlled conditions.	Deliverable: dose-response, pathway-response and safety profiles.
Stage 3	Spatial systems biology	Use single-cell/spatial transcriptomics, proteomics and imaging to test whether spatial patterning is reproduced.	Deliverable: positional-state map.
Stage 4	Small-animal validation	Independent, preregistered studies with blinded assessment and appropriate controls.	Deliverable: reproducibility + safety evidence.
Stage 5	Large-animal translation	Evaluate scale, biomechanics, vascularization, innervation and long-term integration.	Deliverable: translational feasibility package.
Stage 6	Regulatory development	Manufacturing, quality, toxicology, pharmacology/device characterization and regulated clinical design.	Deliverable: regulatory submission package.

Figure 3. Conceptual regenerative microenvironment



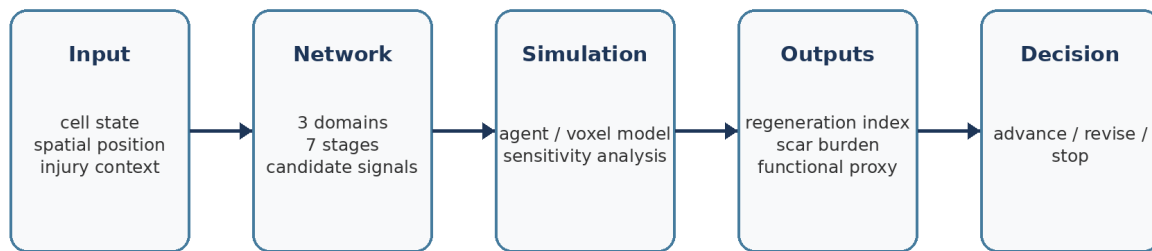
Concept only: no human therapeutic protocol is established by this figure.

Figure 3. Conceptual regenerative microenvironment. This is a research architecture, not a treatment protocol.

6. Computational modelling: making the framework testable

The earlier draft described a detailed 50×50×50 voxel simulation and numerical outcomes. Because the supplied file does not provide the underlying code, parameter-fitting dataset, validation logs, or reproducible repository, those numerical results are removed from the revised manuscript. The stronger scientific position is to specify a model that can be independently implemented and audited.

Figure 4. Computational hypothesis-testing architecture



Purpose: compare hypotheses and identify failure modes before animal studies.

Figure 4. Computational hypothesis-testing architecture.

6.1 Proposed state variables

Each spatial unit can contain: neural/positional state N; immune state I; structural/connective state C; vascular support V; positional identity P; and a growth-control variable G. The model can be extended with extracellular matrix density, oxygenation and mechanical stress.

6.2 Model objectives

The model should estimate regeneration trajectory, spatial fidelity, fibrosis burden, vascular adequacy and growth termination. Parameter inference should use experimentally measured data rather than manually chosen values. Sensitivity analysis should identify parameters that matter most, while uncertainty propagation should show how robust predictions are.

6.3 AI and data science opportunity

A future version can combine mechanistic models with machine learning for parameter inference and candidate ranking. AI should be used as a decision-support layer rather than as a substitute for biological validation. The most valuable dataset would link spatial molecular states to longitudinal morphology and function.

7. Safety, ethics and regulatory boundaries

Regenerative medicine has high translational potential but also high consequence if biological control is inadequate. FDA guidance emphasizes that unapproved cell-based regenerative products can carry risks including infection, unwanted immune responses, ectopic tissue and tumour formation, and that products requiring approval must undergo appropriate regulatory oversight.

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Risk	Why it matters	Design response
Uncontrolled proliferation	Growth pathways can produce unwanted tissue expansion.	Growth-termination layer; long-term monitoring; independent pathology.
Ectopic tissue	Multipotent/progenitor cells may differentiate outside target tissue.	Lineage tracing; spatial containment; release criteria.
Fibrosis	Repair may default to scar formation.	Quantify matrix composition and inflammatory resolution.
Immune complications	Cell products and biomaterials can alter immune state.	Immunophenotyping; cytokine monitoring; dose escalation.
Genomic instability	Cell expansion can introduce abnormal clones.	Genomic QC and predefined exclusion criteria.
Off-target signalling	Pathway modulation can affect distant tissues.	Local delivery where justified; pharmacodynamic monitoring.
Translation gap	Animal regeneration does not imply human regeneration.	Independent replication + large-animal evidence before clinical escalation.

7.1 Human use is not established

Nothing in this manuscript supports self-treatment, off-label cell injections, gene editing, or unsupervised use of retinoids, Hedgehog agonists, growth factors, electrical stimulation or biomaterials. The proposed work belongs within appropriate institutional biosafety, animal ethics, human research ethics and regulatory frameworks.

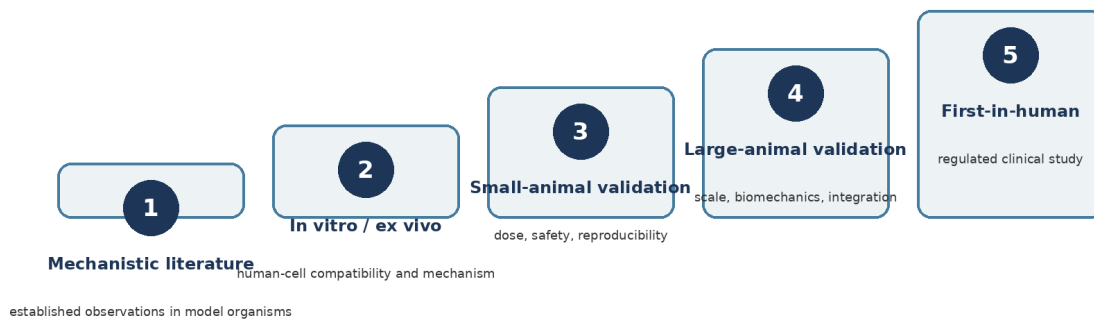
8. Evidence landscape and current maturity

The current literature supports several building blocks of the framework but does not support the claim that complete human limb regeneration is clinically achievable today. Recent reviews emphasize that the field is advancing while translation remains difficult.

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Claim area	Current evidence interpretation
Axolotl regeneration	Strong experimental model for studying complex regeneration and positional information.
Positional memory	Supported by recent molecular and functional studies in axolotl. $\text{NciteÖturn0search4Öturn0search1Ö}$
Hedgehog / VEGF roles	Supported in specific axolotl regeneration contexts; effects are tissue- and stage-dependent. $\text{NciteÖturn1search9Öturn1search15Ö}$
ADSC wound healing	Human clinical evidence suggests potential benefit in selected wounds, but standardization and long-term translation remain issues. $\text{NciteÖturn0search0Öturn0search3Ö}$
Complete mammalian limb regeneration	Not established as a clinical capability.
3–7–12 framework	Novel author-proposed organizing model requiring validation.
Human clinical protocol	Not provided; future work must proceed through regulated development.

Figure 5. Translational evidence ladder



The present manuscript is positioned at levels 1-2: hypothesis generation and research planning.

Figure 5. Translational evidence ladder. The current project is positioned at evidence levels 1–2, not at clinical efficacy.

9. Scientific differentiation and intellectual-property strategy

The strongest defensible contribution is not the generic idea that animals regenerate. The differentiation should be the structured architecture that converts comparative biology into a measurable, computationally testable and safety-gated research platform.

Potential asset	Possible protection / value
3–7–12 systems architecture	Methodology / software / know-how, subject to legal review and prior-art search
Computational model	Software, model implementation, parameter-learning pipeline
Spatial biomarker panel	Potential diagnostic / research-use IP if novel and validated
Biomaterial + delivery system	Potential patentable composition or device if novel
Combination strategy	Potential method-of-use IP if experimentally supported
Dataset	Proprietary dataset and model-training advantage
Research platform brand	Trademark / commercial identity subject to clearance

Before public disclosure of patentable implementation details, the author should obtain professional IP advice and perform a formal prior-art search. A DOI can establish a public timestamp for authorship and disclosure, but it does not by itself create patent rights.

10. Industry and investment proposition

For biotechnology companies, medical-device developers, tissue-engineering groups, pharmaceutical companies and strategic investors, the project is best presented as an enabling research platform rather than as a finished regenerative therapy.

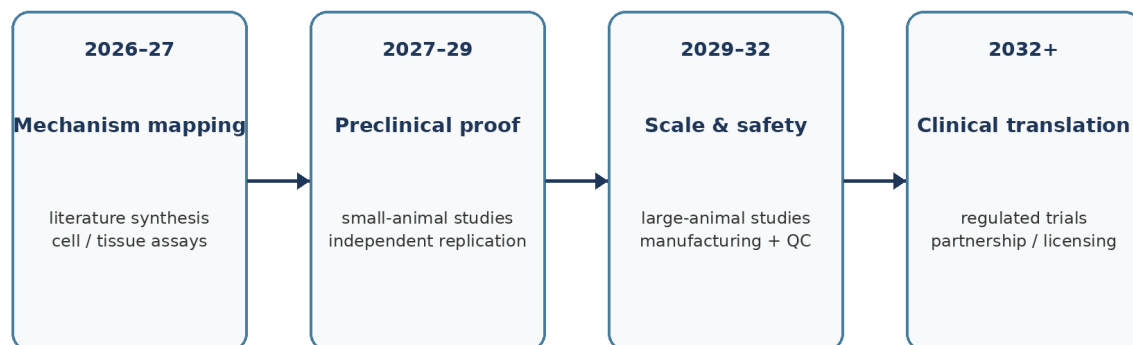
Partner type	Potential collaboration
Regenerative medicine company	Joint preclinical validation and biomaterial/cell platform development
Pharmaceutical / biotech	Pathway-modulation screening and delivery systems
Medical-device company	Scaffold, electrical stimulation or controlled local-delivery technologies
AI / computational biology company	Mechanistic digital-twin and candidate-ranking platform
Academic medical center	Independent validation, imaging, omics and regulated translational studies
Strategic investor	Fund staged R&D milestones in exchange for defined IP / licensing rights
Contract research organization	GLP-aligned safety, toxicology and large-animal studies

10.1 Investment thesis — disciplined version

The investment case is based on the possibility that regeneration can be treated as a controllable systems-engineering problem rather than a single-target drug problem. The near-term value proposition is therefore research infrastructure, proprietary modelling, datasets, biomaterials and validated preclinical combinations. The long-term upside would come from successful validation and licensing or co-development of specific therapeutic implementations.

No return, valuation, clinical success rate or timeline is guaranteed. Capital should be released through milestone-based financing: model validation → human-cell evidence → independent animal replication → large-animal evidence → regulatory development.

Figure 6. Research-to-industry pathway



Industry proposition: a structured research platform, not a guaranteed clinical product.

Figure 6. Illustrative research-to-industry pathway. Dates are planning windows, not guaranteed achievements.

11. Author and collaboration profile

Vahid Nezamivand Chegane is an independent interdisciplinary researcher and founder of the AquaNexus Project. His work emphasizes systems thinking across water, energy, environmental resilience and emerging technology concepts. The present manuscript extends that systems-oriented approach into regenerative medicine by proposing a structured framework for comparative biology, computational modelling and translational research.

The author's role in this project is best described as concept originator and systems architect. Experimental claims should be established through independent laboratory partnerships, preregistered protocols, appropriate ethics approvals and transparent data sharing. This division of roles is intentionally designed to make the project easier for universities and companies to diligence.

Author	Vahid Nezamivand Chegane
Role	Independent Researcher; Concept Originator / Systems Architect
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Project association	AquaNexus Project
Contact	vahidmaykoltarh@yahoo.com
Collaboration formats	Academic collaboration • co-development • licensing • sponsored research • strategic investment

12. Conclusion

Animal regeneration research has moved from descriptive zoology toward a mechanistic science of positional information, cell-state control, immune regulation, vascular support and tissue-scale patterning. Recent work in axolotl regeneration is particularly valuable because it shows that cells can retain spatial identity and coordinate signalling after injury.

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The 3–7–12 framework proposed by Vahid Nezamivand Chegane is positioned here as a hypothesis-generating systems architecture. Its central idea is that regenerative outcomes should be evaluated through coordinated domains, sequential functional states and interacting signalling modules, with explicit growth-control and safety layers.

The revised manuscript intentionally does not claim completed rat validation or a ready human limb-regeneration protocol. Instead, it defines the experiments required to determine whether the framework has predictive and translational value. This change makes the work more scientifically defensible and more suitable for discussions with universities, biotechnology companies, medical-device developers and strategic investors.

The next meaningful milestone is not a larger claim. It is independent evidence: a reproducible computational model, human-cell experiments, spatial measurements, preregistered animal studies, and transparent safety endpoints. If those milestones are achieved, the 3–7–12 architecture could become a platform for a new generation of regenerative engineering research.

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Appendix A. Major corrections applied to the supplied manuscript

Original draft element	Revision
'First experimental evidence' and rat outcomes	Removed as completed findings because the supplied document does not provide auditable raw data, protocol, ethics documentation or independent replication.
Rigid statement that exactly twelve pathways govern regeneration	Reframed as a 12-module candidate network.
90–120 day complete human limb regeneration prediction	Removed as an unsupported clinical prediction.
Specific drug / dose / temperature clinical-style protocol	Removed; replaced with research-stage hypotheses and safety gates.
Cost estimate for clinical application	Removed from scientific claims; commercialization is framed around staged R&D.
2030/2035/2040/2050 definitive clinical milestones	Replaced with milestone-based translational pathway rather than guaranteed dates.
'3–7–12 codes govern regeneration'	Rewritten as '3–7–12 is an author-proposed conceptual framework requiring validation.'
Copyright / collaboration language	Retained as a practical IP notice, but clarified that legal protection should be assessed professionally.

This appendix is included to make the revision auditable. It distinguishes scientific correction from stylistic rewriting and is intended to help reviewers and prospective partners understand why the revised manuscript is more conservative than the original.

Appendix B. Suggested data package for future credibility

- Preregistered study protocol and statistical analysis plan.
- Animal ethics approval and reporting according to ARRIVE guidance.
- Raw imaging files, blinded scoring sheets and analysis scripts.
- Cell identity, viability, genomic stability and sterility/QC records.
- Full simulation code with parameter definitions, training data and sensitivity analysis.
- Independent replication by a laboratory with no financial dependence on the author.
- Predefined stopping rules for toxicity, ectopic growth and failure of regeneration.
- Clear separation between exploratory findings and confirmatory endpoints.